2013 JC2 Prelim P3 Answers for Exchange

1 Hydrolysis involves the cleavage of chemical bonds on reaction with water. It is often used in the synthesis of important chemical intermediates.

(a) The rate of the hydrolysis of bromoalkanes with aqueous potassium hydroxide depends upon the chemical structure of the bromoalkane.

(i) Explain what is meant by the ***rate*** of a chemical reaction.

The change in the concentration of a product or reactant with time.

The table below shows the results of a series of hydrolysis experiments that was carried out using different concentrations of the bromoalkane, C_4H_9Br, and aqueous KOH.

experiment	***initial* [C_4H_9Br] / mol dm^{-3}**	***initial* [KOH] / mol dm^{-3}**	***initial* rate/ mol dm^{-3} s^{-1}**
I	0.10	0.10	0.024
II	0.15	0.10	0.036
III	0.20	0.20	0.048

(ii) Use the given data to deduce the order of reaction with respect to each of the two reactants in the hydrolysis reaction, showing how you arrive at your answers.

rate = $k[C_4H_9Br]^m[KOH]^n$
Exp II / Exp I: $(0.036/0.024) = (0.15/0.10)^m$ ➔ $m = 1$
Exp III / Exp I: $(0.048/0.024) = (0.20/0.1)^1(0.20/0.10)^n$ ➔ $n = 0$

(iii) Hence, write the rate equation for the reaction and state the units of the rate constant.

rate = $k[C_4H_9Br]^1$, units for k = s^{-1}

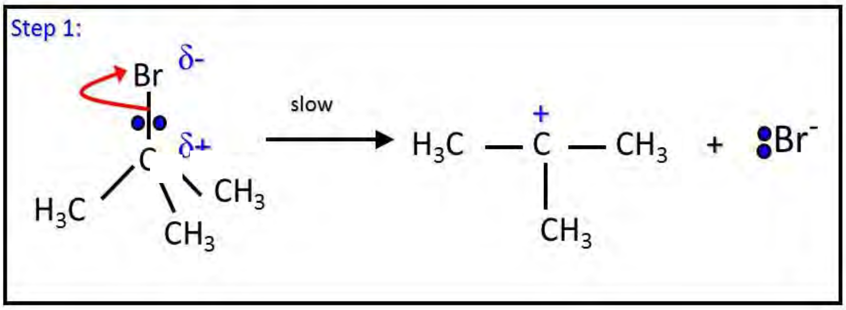
(iv) Given that the bromoalkane is achiral, use your answer in **(a)(iii)** to suggest a possible structural formula for the bromoalkane.

Hence, propose a mechanism for the hydrolysis of the bromoalkane in aqueous potassium hydroxide that is consistent with the rate equation.

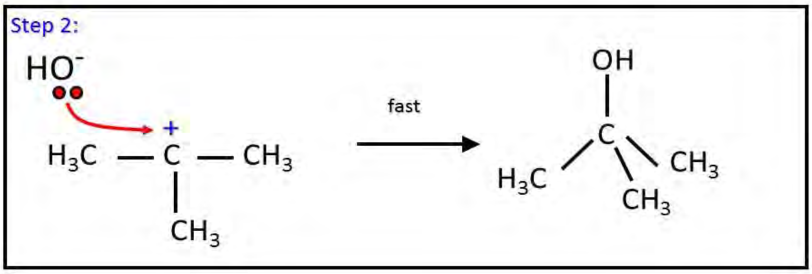
Nucleophilic substitution (S_N1)

$(CH_3)_3CBr$

Dissociation of the alkyl bromide in a slow step to generate a carbocation intermediate and a Br- ion



The nucleophile OH- attacks the carbocation intermediate in a fast step to form the neutral alcohol product



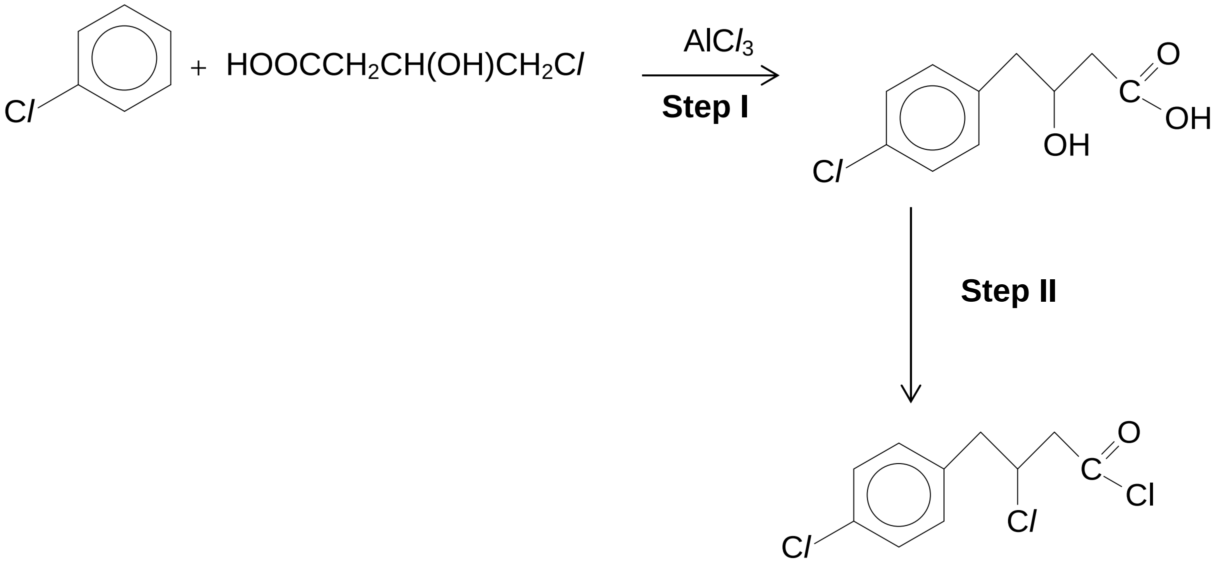
(v) How would you expect the rate of hydrolysis to change if the bromoalkane was replaced by an iodoalkane of similar structure? Explain your answer with the use of suitable values from the *Data Booklet*.

The rate will increase.
The bond energy of C-I *(240 kJ/mol)* is smaller than C-Br *(280 kJ/mol).* Hence, the C-I bond is easier to break.

(vi) In a separate experiment, the bromoalkane from **(a)(iv)** is treated with alcoholic KOH. State the type of reaction that occurs and give the structural formula of the organic product formed.

Elimination
$(CH_3)_2C{=}CH_2$ [12]

(b) Consider the reaction scheme below.



compound **T**

(i) State the type of reaction in **Step I.**

Electrophillic substitution

(ii) State the reagent(s) and condition(s) required for **Step II.**

PCl_3 or PCl_5 or $SOCl_2$, r.t.p

(iii) Explain the difference in the reactivities of the 3 chlorine atoms in compound **T.**

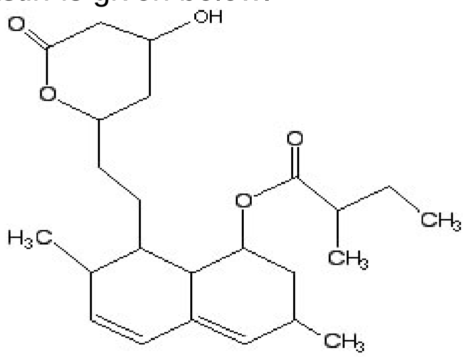
C*l* bonded to the benzene ring is the most unreactive due to the overlapping of its p-orbital with π-electron cloud of the benzene ring. (Or lone pair of electrons on Cl is delocalized into the benzene ring.)
Hence, the C-C*l* bond is very strong and not easily broken.

-The $\delta+$ C of the acyl chloride is bonded to two electronegative atoms (– C*l* & O) while the $\delta+$ C of chloroalkane is bonded to only 1 electronegative atom (-Cl).

- Acyl chloride is most susceptible to nucleophilic attack because of the relatively bigger partial positive charge on the C atom compared to the C atom of the chloroalkane.

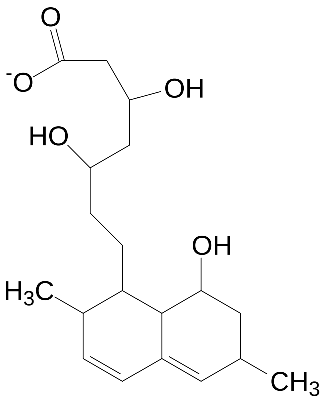
[5]

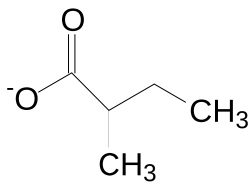
(c) The structure of *lovastin* is given below.



lovastin

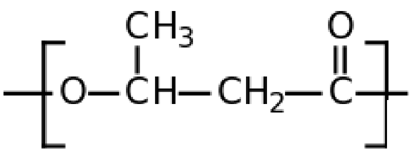
Draw the structure of the products formed when *lovastin* reacts with hot, aqueous sodium hydroxide.



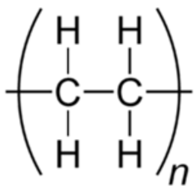


[2]

(d) The structures below show small sections of the polymers – *polyhydroxybutyrate* and *polyethene*. Both polymers are used in the manufacture of films for packaging.



Polyhydroxybutyrate



polyethene

(n > 500)

Unlike polyethene, *polyhydroxybutyrate* degrades within 4 weeks in landfills. Suggest an explanation for this.

polyhydroxybutyrate comprises ester links which can be hydrolysed / broken with weak acids or alkalis. [1]

[Total: 20]

2 This question is about magnesium and its compounds.

(a) Magnesium hydroxide can be used as a fire-retardant as it is able to decompose upon heating at 332°C according to the following equation:

$$Mg(OH)_2\ (s) \xrightarrow{\Delta H_r} H_2O\ (g) + MgO\ (s)$$

	Enthalpy change / kJ mol^{-1}
Standard enthalpy change of atomisation of magnesium	147
1st electron affinity of oxygen	-142
2nd electron affinity of oxygen	844
Lattice energy of magnesium oxide	-3920
Standard enthalpy change of formation of water	-289
Latent heat of vaporization of water	22.3
Standard enthalpy change of formation of magnesium hydroxide	-927

(i) Explain the difference between the first and second ionisation energies of magnesium.

The second ionisation energy is more endothermic than the first.

The first electron is removed from a neutral atom whereas the second electron is removed from a positively charged ion. The latter has greater attraction for the valence electrons so more energy required to remove the next electron.

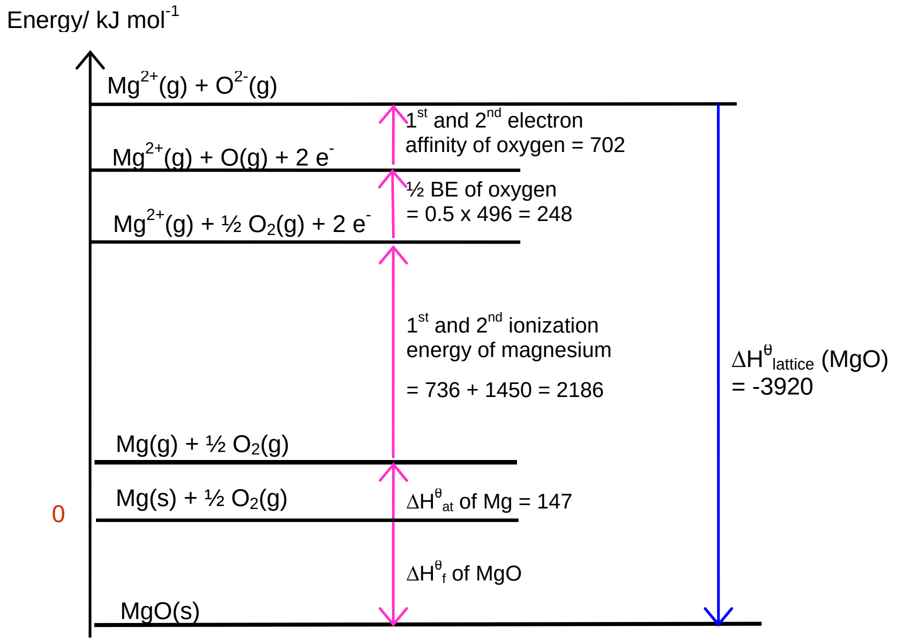
<u>OR</u>

Nuclear charge remains same whilst no. of electrons decreases. Remaining electrons held more tightly, more energy required to remove the next electron.

(ii) Suggest why the second electron affinity of oxygen is endothermic.

Repulsion occurs as the next electron is added to an ion which is already negatively charged.

(iii) Using relevant values from the table above and the *Data Booklet*, draw a fully labeled energy level diagram to find the standard enthalpy change of formation of magnesium oxide.



ΔH^{θ}_{f} of MgO $= 147 + 2186 + 248 + 702 - 3920$

$= -637$ kJ mol^{-1}

(iv) Using your answer in **(a)(iii)** and relevant values provided in the table above, calculate $\Delta H_{r,}$ enthalpy change for the thermal decomposition of magnesium hydroxide. State any assumptions made in your calculations.

Assuming standard enthalpy change of formation of water, magnesium oxide and magnesium hydroxide are constant with temperature.

$\Delta H_r = -637 - 289 + 22.3 - (-927) = 23.3$ kJ mol^{-1}

(v) By considering your answer in **(a)(iv)**, suggest how magnesium hydroxide functions as a fire-retardant.

Endothermic thermal decomposition reaction absorbs energy from the fire.

(vi) Barium hydroxide undergoes a thermal decomposition reaction similar to that of magnesium hydroxide.

By quoting suitable data from the Data Booklet, deduce whether barium hydroxide will decompose at a higher or lower temperature than magnesium hydroxide. Explain your answer clearly.

Ionic radii of Mg^{2+} : 0.065nm ; Ba^{2+} : 0.135nm

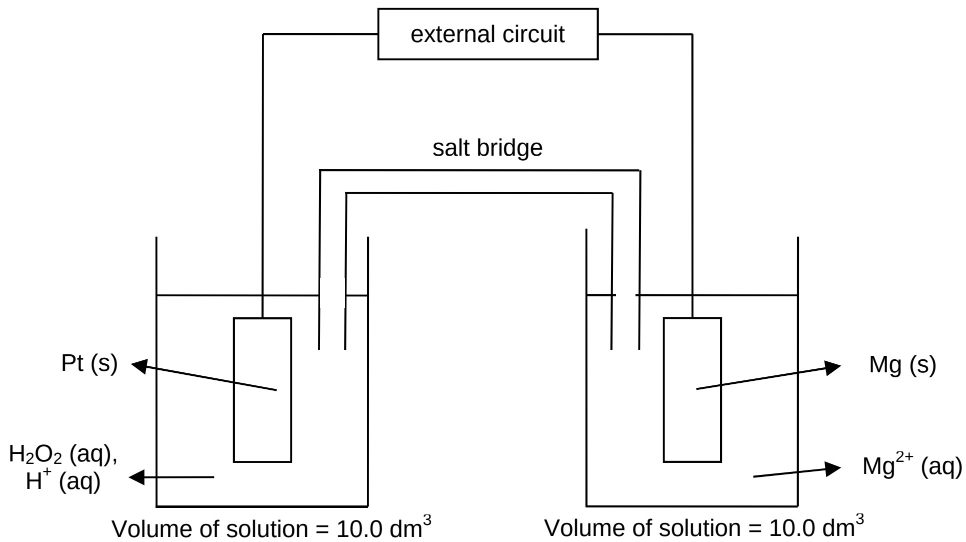
(Same charge but) Larger radii of Ba^{2+} compared to Mg^{2+}, thus lower charge density. Ba^{2+} has lower polarising power, leading to less distortion of the hydroxide electron cloud. (The O-H bond in barium hydroxide is weakened to a smaller extent than in magnesium hydroxide.)

Thus, barium hydroxide will decompose at a higher temperature than magnesium hydroxide.

[13]

(b) A magnesium-hydrogen peroxide fuel cell has been developed to power undersea vehicles.

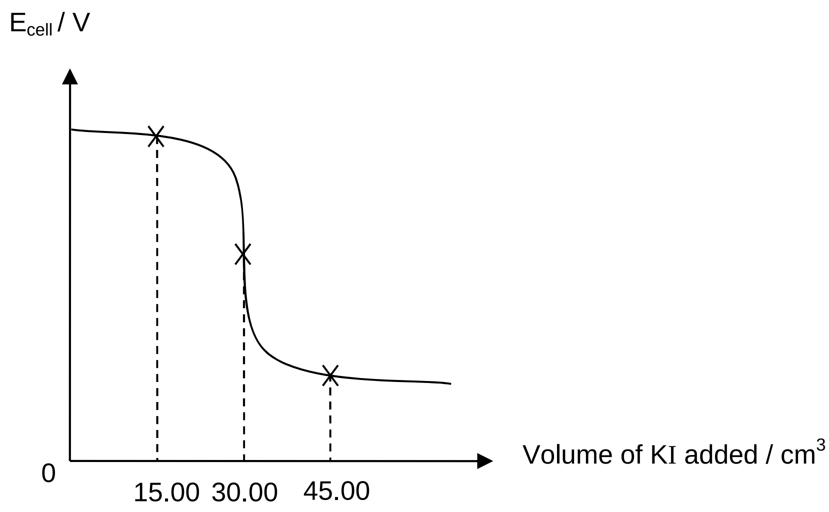
The following diagram is a simplified representation of the cell.



The energy derived from this fuel cell depends on the concentration of H_2O_2 used.

A 25.0 cm^3 sample of the aqueous hydrogen peroxide solution was taken from the cell and its concentration was estimated by titration with 1.50 mol dm^{-3} aqueous potassium iodide.

The E_{cell} was measured against a standard hydrogen electrode as aqueous potassium iodide was added. The following graph was obtained:



(i) Write a balanced equation for the reaction between potassium iodide and acidified H_2O_2.

$H_2O_2 + 2H^+ + 2\,I^- \rightarrow 2H_2O + I_2$

(ii) Calculate the concentration of hydrogen peroxide in the 25.0 cm^3 of sample.

Equivalence volume of KI = 30.00 cm^3

Amount of KI = 1.5 x 30 /1000 = $\underline{0.0450\text{ mol}}$

$2KI \equiv H_2O_2$

Amount of H_2O_2 = 0.0450 / 2 = 0.0225 mol

Concentration of H_2O_2 = (0.0225) / (25/ 1000) = $\underline{0.900\text{ mol dm}^{-3}}$

(iii) Determine the minimum mass of Mg metal required at the anode in the fuel cell to fully utilise the hydrogen peroxide present in the cell shown above.

Amount of H_2O_2 in set-up = 0.900 x 10.0 = 9.00 mol

$Mg \equiv H_2O_2$

Mass of Mg required = 9.00 x 24.3 = 219 g (3.s.f.)

(iv) Suggest why calcium is not a suitable replacement for magnesium in this fuel cell set-up.

Calcium will react with water so it cannot be used as a fuel in this set-up.

(v) Fuel cells are increasingly being used to produce energy.

Suggest two environmental disadvantages of using the conventional internal combustion engine found in cars compared to the magnesium-hydrogen peroxide fuel cell.

- Car engine produces air pollutants such as
  - Carbon monoxide from incomplete combustion: prevents haemoglobin in the blood from transporting oxygen
  - Oxides of nitrogen or sulfur dioxide: Respiratory irritant leading to respiratory tract infections
    or leading to acid rain, ozone depletion for oxides of N.

- Car engine produces greenhouse gases e.g. carbon dioxide
  - Leading to global warming

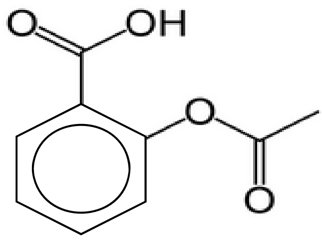
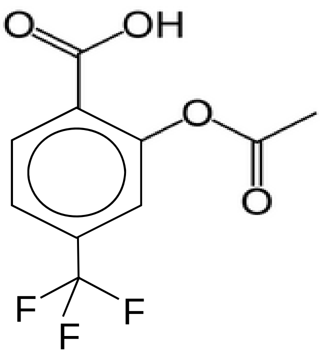
*Any two points

[7]

[Total: 20]

3 Pharmaceutical drugs such as aspirin and triflusal, have been found to prevent heart attacks or strokes. The active ingredients in aspirin and triflusal are the monobasic acids, $\mathbf{HA_1}$ and $\mathbf{HA_2}$, respectively.

The pKa values of both acids are shown in the table below.

Compound	**pKa**
 **HA_1**	4.80
 **HA_2**	3.00

(a) **(i)** State and explain how the acidities of compounds $\mathbf{HA_1}$ and $\mathbf{HA_2}$ compare with each other.

HA_2 is more acidic than HA_1 as it has a lower pKa value.

HA_2 has an additional electron withdrawing $–CF_3$ group which stabilizes its corresponding anion (or conjugate base) by dispersing the negative charge on the anion.

The anion of HA_2 is more stable than the anion of HA_1. Hence, HA_2 has a greater extent of dissociation than HA_1.

(ii) Calculate the pH of a 0.20 mol dm^{-3} solution of compound $\mathbf{HA_1}$.

$Ka = [H^+][A_1^-] / [HA_1]$

$= [H^+]^2 / 0.20$; assume acid dissociation is negligible & $[H^+] = [A^-]$

$10^{-4.8} = [H^+]^2 / 0.20$

$[H^+] = 1.78 \times 10^{-3}$

pH = 2.75 [5]

Aspirin has been found to cause stomach irritation in some people. In such cases, alternative forms of the drug have been prescribed. These alternative forms contain a buffer system of $\mathbf{HA_1}$ and its sodium salt.

(b) **(i)** Explain what is meant by a buffer solution.

A buffer solution is one that resists changes in pH when a small amount of acid or base is added.

(ii) Calculate the pH of a buffer solution containing 20.00 cm^3 of 0.20 mol dm^{-3} $\mathbf{HA_1}$ mixed with 20.00 cm^3 of 0.25 mol dm^{-3} of its sodium salt.

[Salt] = 0.25 x (20/40) = 0.125 mol dm^{-3}

$[HA_1]$ = 0.20 x (20/40) = 0.100 mol dm^{-3}

$pH = pKa + \lg[salt]/[HA_1]$

$= 4.8 + \lg(\mathbf{0.125 / 0.100})$

$= \mathbf{4.90}$

(iii) Calculate the pH when 5.00 cm^3 of 0.10 mol dm^{-3} HCl is added to the buffer solution in b(ii).

$A_1^- + HCl \rightarrow HA_1 + Cl^-$

[salt] new = [(0.25 x 0.02) – (0.005 x 0.10)] / 0.045 =0.100

[acid] new = [(0.20 x 0.02) + (0.005 x 0.10)] / 0.045= 0.100 [5]

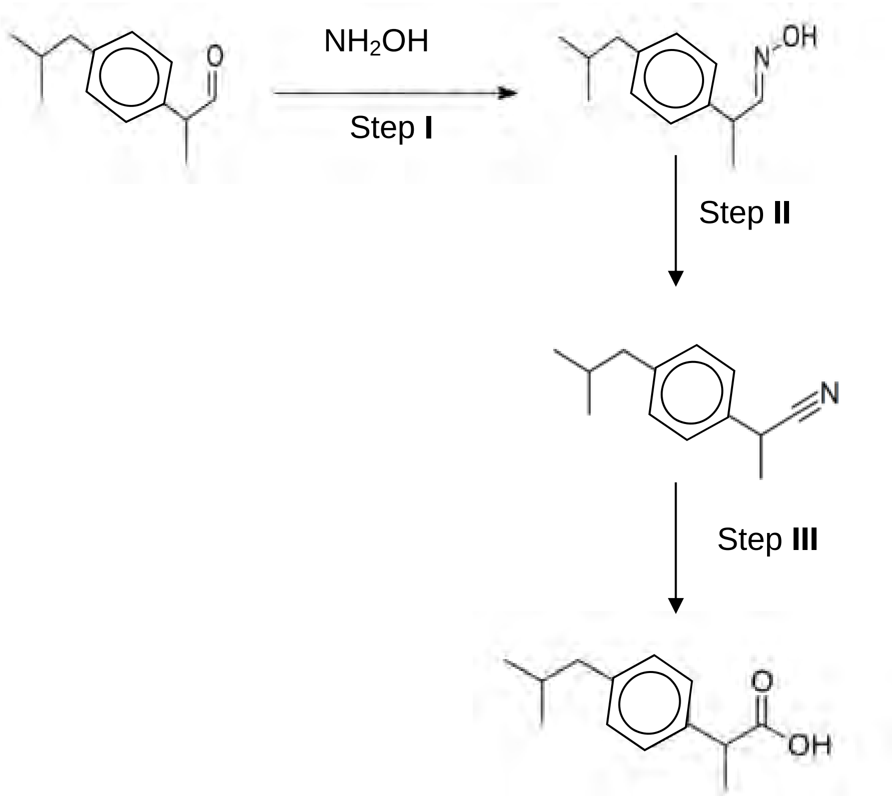
pH = pKa + lg [salt]/[acid]

= 4.8 + 0

= **4.8**

(c) Ibuprofen is a pharmaceutical drug that is manufactured for a similar purpose as aspirin, which is to relieve minor aches and pains.

Ibuprofen can be synthesised according to the following reaction scheme.



Ibuprofen

(i) Suggest the type of reaction in step **I** and step **II**.

Step I : condensation or addition-elimination

Step II: elimination

(ii) Suggest the reagents and conditions for step III.

H_2SO_4 (aq) or HCl (aq) , heat or reflux

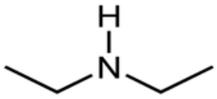
(iii) Ibuprofen is sparingly soluble in water. Suggest an inorganic reagent that could be added to Ibuprofen to increase its solubility.

Any reactive metal (Group I or Group II) or base.

[4]

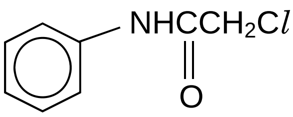
(d) Compound **C** is used in the synthesis of a common local anesthetic drug, **D**.

When compound **C** (C_8H_8NOCl) is reacted with hot aqueous sodium hydroxide, compound **E** (C_6H_7N) and a sodium salt of acid **F** are formed. **E** reacts with chlorine water to give a tri-substituted organic product.

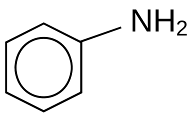
C reacts with alcoholic  under pressure, in the last stage of synthesis of the drug, to give **D**.

Deduce the structures of **C to F,** explaining the reactions that occur. [6]

Structures C - F :



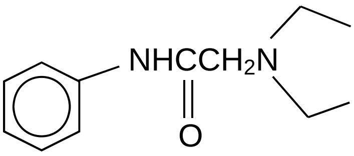
C



E

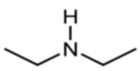
$HOCH_2COOH$

F



D

Explanations:

Evidence	Deduction
C (C_8H_8NOCl) is reacted with hot aqueous sodium hydroxide	C undergoes base hydrolysis with NaOH (aq) C contains an amide functional group. (phenylamine and $HO\text{-}CH_2COO^-Na^+$ are formed.)
E reacts with chlorine water	E undergoes electrophilic substitution with chlorine water to give $C_6H_4Cl_3N$; E is phenylamine.
C reacts with alcoholic  under pressure	C undergoes nucleophilic substitution with diethylamine to give D ; C must contain a halogenoalkane functional group.

4 The transition elements such as iron and cobalt and their compounds have an important role in industry as catalysts.

(a) The reaction between iodide and peroxodisulfate ions is an example of a reaction that is catalysed by iron(II) ions, Fe^{2+}.

$$2\,I^-(aq) + S_2O_8^{\,2-}(aq) \rightarrow I_2(aq) + 2\,SO_4^{\,2-}(aq)$$

(i) The role of iron(II) ions here is described as a ***homogeneous catalyst.*** Explain what is meant by the term *homogeneous catalyst.*

A homogeneous catalyst is a substance which is in the same phase (or state) as the reactants and increases the reaction rate while itself remaining chemically unchanged at the end of the reaction.

(ii) Suggest a reason why the reaction between iodide and peroxodisulfate ions requires a catalyst.

As both iodide and peroxodisulfate ions are anions, repulsion between the anions will cause the activation energy to be high, leading to a slow reaction rate.

(iii) Besides iron(II) ions, cobalt(II) ions are also suitable catalysts for the above reaction.
With reference to relevant $E^\ominus$ values in the *Data Booklet*, describe how Co^{2+} ions are able to catalyse the reaction between iodide and peroxodisulfate. Write equations where appropriate.

Using the following Data from the Data Booklet,

$I_2 + 2e^-$	$\rightleftharpoons 2I^-$	$E^\ominus = +0.54$ V
$S_2O_8^{2-} + 2e^-$	$\rightleftharpoons 2SO_4^{2-}$	$E^\ominus = +2.01$ V
$Co^{3+} + e^-$	$\rightleftharpoons Co^{2+}$	$E^\ominus = +1.82$ V

Co^{2+} first reduces $S_2O_8^{2-}$ while itself is oxidised to Co^{3+}:
$2Co^{2+} + S_2O_8^{2-} \rightarrow 2Co^{3+} + 2SO_4^{2-}$

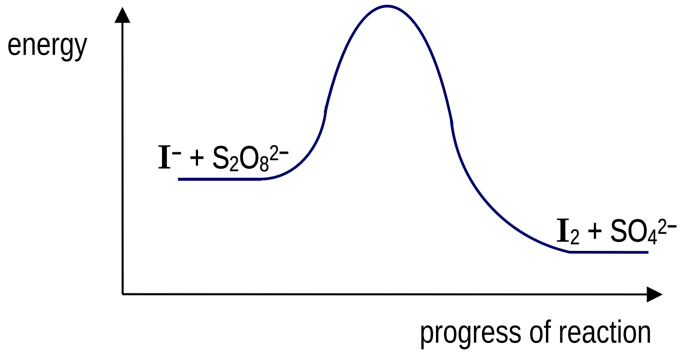
$E^\ominus_{cell} = +2.01 - (+1.82) = +0.19V > 0 \Rightarrow$ reaction is feasible.

Co^{3+} then oxidises I^- and is itself reduced to give Co^{2+}:

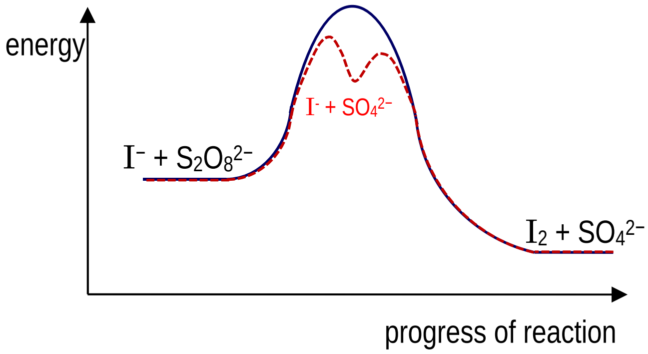
$2Co^{3+} + 2I^- \rightarrow \underbrace{2Co^{2+}}_{\text{(regenerated)}} + I_2$

$E^\ominus_{cell} = +1.82 - (+0.54) = +1.28V > 0 \Rightarrow$ reaction is feasible.

(iv) The reaction pathway diagram for the uncatalysed reaction is given below.

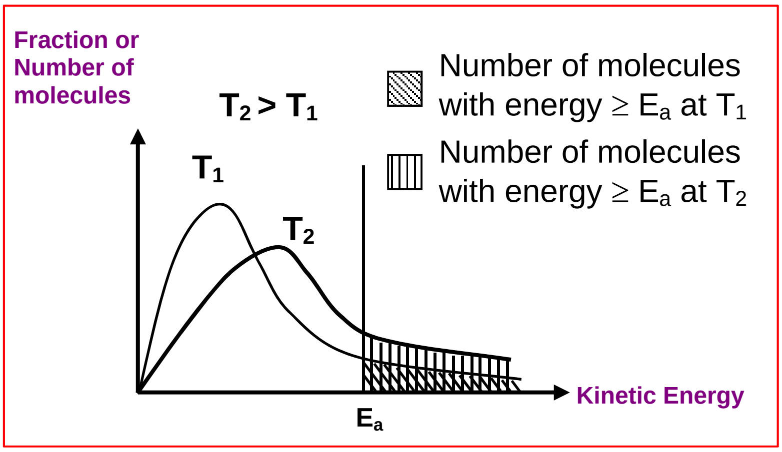


Copy the diagram and draw, on the same axes, a possible reaction pathway diagram for the catalysed reaction.



[7]

(b) The rate of a reaction can also be increased by increasing the temperature. By means of the Maxwell-Boltzmann distribution of molecular energies, explain how the increase in temperature leads to an increase in reaction rate.



[3]

When temperature increases, kinetic energy of reactant particles increases.

The Boltzmann distribution shows that the number of reactant particles with energy greater than or equal to E_a increases.

Frequency of <u>effective</u> collisions increases.

Hence, rate of reaction increases.

(c) Another characteristic feature of the chemistry of transition elements is their tendency to form complexes.
When aqueous cobalt(III) chloride reacts with ammonia under various conditions, compounds containing different cobalt complexes can be formed.

Two of such compounds containing the cobalt complexes, **X** and **Y**, have the same formula $CoCl_3(NH_3)_4$. A third compound, **Z**, has the formula $CoCl_3(NH_3)_5$.

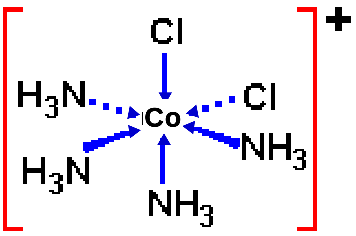
- When treated with aqueous silver nitrate, both **X** and **Y** each form 1 mole of a white precipitate per mole of the compound whereas **Z** forms 2 moles of the white precipitate.
- Although the cobalt complex ion in **X** and **Y** have the same formula, the complex ion in **X** does not have a dipole moment whereas the ion in **Y** has a dipole moment.

(i) Identify the white precipitate that forms when **X, Y** and **Z** are treated with aqueous silver nitrate. Answer: AgCl

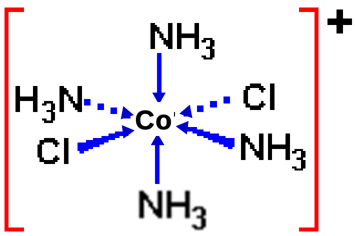
(ii) Suggest the formulae of the cobalt complex ions in **X** and in **Z**.

X : $[Co(NH_3)_4(Cl)_2]^+$ **Z** : $[Co(NH_3)_5(Cl)]^{2+}$

(iii) Suggest a structure for the cobalt complex ion in **X** and in **Y** which may be used to explain the presence or absence of a dipole moment.



Y



X

[5]

(d) Besides the use of transition elements and their compounds as catalysts, biological catalysts such as enzymes are also used in industries such as in food production. Coagulating agents used in the manufacture of cheese include the enzyme *Chymosin*.

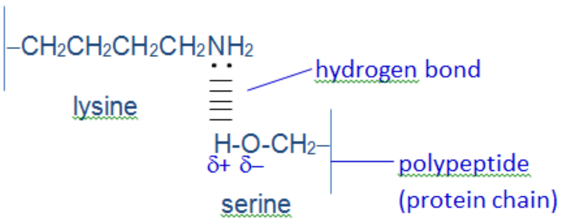
Chymosin is a protein molecule comprising more than 300 amino acids. Some of the amino acids found in *Chymosin* are shown below.

amino acid	formula of side chain (R in $RCH(NH_2)CO_2H$)
aspartic acid	$–CH_2COOH$
leucine	$–CH_2CH(CH_3)_2$
serine	$–CH_2OH$
threonine	$–CH(OH)CH_3$
valine	$–CH(CH_3)_2$
lysine	$–CH_2CH_2CH_2CH_2NH_2$

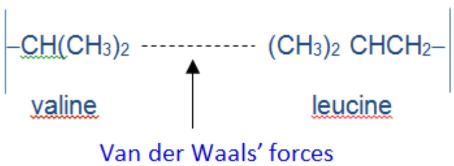
(i) With reference to relevant amino acids from the table above, describe **two** forms of interaction that might be responsible for maintaining the tertiary structure of *Chymosin*. Illustrate your answer with suitable pairs of amino acids from the table above.

Any 2 of the following:

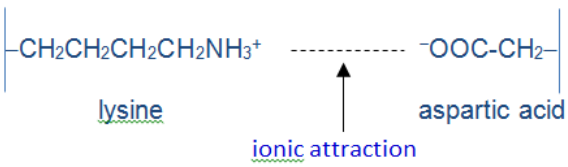
1) hydrogen bonding between R groups of lysine (or asp acid) and serine (or threonine)



2) Van der Waals' forces between R groups of valine and leucine



3) ionic bonding between R groups of aspartic acid and lysine



(ii) It has been found that *Chymosin* loses its activity under conditions of extreme pH.
With reference to the amino acid residues present in *Chymosin*, explain how conditions of extreme **high** pH leads to this loss of activity.

Extreme high pH changes deprotonate the ionic R groups such as $–CH_2CH_2CH_2CH_2NH_3^+$ and so destroy the ionic bonds (interactions) which help to stabilise the tertiary structure.

As a result, the protein structure is altered (protein is denatured) and the enzyme loses its activity.

[5]

5 Chlorine is an important element for the chemical industry. In nature, chlorine is seldom found in its elemental form, but mostly as chlorides and chlorine oxides.

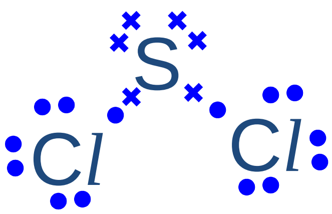
(a) Describe and explain what you would see when solid sodium chloride and solid phosphorus pentachloride are separately added to an aqueous solution containing a few drops of Universal Indicator.

Write an equation for any reaction that occurs. [3]

- Solid Na*Cl* dissolves in water to give a neutral solution, and the Universal Indicator remains green.
- Solid PCl_5 reacts/ hydrolyse [not dissolve] in water to produce an acidic solution of HCl and/or H_3PO_4, which turns the Universal Indicator red.
- $PCl_5 + 4\ H_2O \longrightarrow H_3PO_4 + 5\ HCl$

(b) Sulfur dichloride, SCl_2, is formed from the chlorination of elemental sulfur. It is a cherry-red liquid at room temperature and pressure, and has a boiling point of 59°C.

(i) Draw a dot-and-cross diagram showing the valence electrons in SCl_2, and use the VSEPR Theory to predict the shape of SCl_2.



- There are 2 bond pairs and 2 lone pairs of electrons around the central S atom. By VSEPR Theory, the electron pairs will arrange themselves as far apart as possible. Hence, the shape of SCl_2 is bent.

(ii) Assuming ideal gas behaviour, determine the volume occupied by a sample of 1.50 g of SCl_2 at 80°C and 1 atm.

Using the general gas equation, pV = nRT,

$$(1.01\times10^5)V = \frac{1.50}{103.1}(8.31)(273+80)$$

$\Rightarrow V = 4.23 \times 10^{-4}\ m^3$ (or = **423** cm^3)

(iii) The actual volume occupied by the 1.50 g sample of SCl_2 is smaller than that determined in **b(ii)**. Suggest an explanation for this observation. [5]

- The calculation in (b)(ii) is based on ideal gases having negligible intermolecular interaction. However, there is significant attraction between SCl_2 molecules, which results in the molecules being held closer together, resulting in a smaller volume.

(c) Sulfur dichloride can further react with sulfur dioxide and chlorine to form thionyl chloride, $SOCl_2$, according to the equation:

$SCl_2(g) + SO_2(g) + Cl_2(g) \rightleftharpoons 2\ SOCl_2(g)$ *Reaction I*

When an equimolar mixture of SO_2, Cl_2, and SCl_2 at an initial total pressure of 108 atm was allowed to reach equilibrium at 80°C, the partial pressure of $SOCl_2$ is found to be 55.2 atm.

(i) Write an expression for K_p for *Reaction I*, stating its units.

$$K_p = \frac{(p_{SOCl_2})^2}{(p_{SCl_2})(p_{SO_2})(p_{Cl_2})}\ \text{atm}^{-1}$$

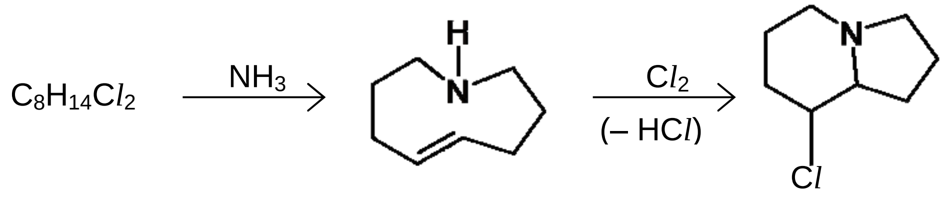
(ii) Determine the equilibrium partial pressure of SO_2, Cl_2, and SCl_2.

	$SCl_2(g)$ +	$SO_2(g)$ +	$Cl_2(g)$ ⇌	$2\ SOCl_2(g)$
Initial pressure /atm	$\frac{108}{3}=36$	$\frac{108}{3}=36$	$\frac{108}{3}=36$	0
Change in pressure /atm	-½ (55.2)	-½ (55.2)	-½ (55.2)	+ 55.2
Pressure at Eqlm /atm	8.4	8.4	8.4	55.2

(iii) Hence, determine a value for the equilibrium constant, K_p of the reaction. [5]

At 500K, $K_p = \dfrac{55.2^2}{(8.4)(8.4)(8.4)}$ **= 5.14 atm^{-1}**

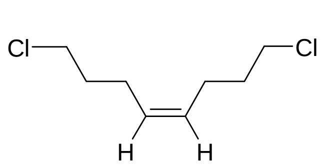
(d) Chlorine is used extensively in organic addition and substitution reactions. The following reaction scheme shows how an aliphatic compound **A**, $C_8H_{14}Cl_2$, can be converted to the monocyclic compound **B**, and subsequently to the bicyclic product **C**.



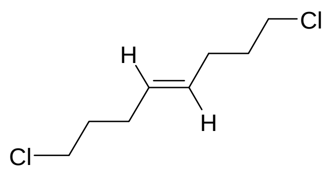
A **B** **C**

(i) Compound **A** exhibits geometric isomerism. Draw and label clearly the isomers of compound **A**.

<u>cis isomer</u>



<u>trans isomer</u>



(ii) State and explain the difference in basic strength between ammonia and compound **B**.

- In compound **B**, there are 2 electron donating alkyl groups that increases the electron density on the N atom.
- This causes the lone pair of electron on N atom (of compound **B**) to be more readily available on accepting a proton.
- Hence, compound **B** is a stronger base than ammonia.

(iii) The reaction of compound **B** with chlorine gas begins with the electrophilic attack of chlorine on the alkene functional group. Describe the two-step mechanism for formation of the bicyclic product **C** from compound **B**. [7]